Sticky Expectations in Durable Goods: Measurement and Consequences

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Abstract

This paper tests whether sticky consumer beliefs about durable-goods buying conditions distort actual durable spending beyond what macroeconomic fundamentals explain. Using the University of Michigan Index of Buying Conditions for Large Household Durables as the subjective belief, I build a rational benchmark by training a rolling-window LASSO on twenty-one real-time macroeconomic predictors to forecast real PCE durable goods growth, adapting the approach of Du, Monninger, Qiu, and Wang (2025). A first-stage underreaction test rejects rational expectations decisively: subjective beliefs adjust by only 44–60% of what fundamentals warrant, a degree of stickiness close to what Du et al. find for unemployment risk. The downstream spending consequences are weaker. A symmetric predictive regression finds no significant effect of beliefs on future durable spending, and the asymmetric specification yields a significant short-horizon effect only in shock-trimmed samples, with optimism and pessimism deviations roughly symmetric rather than pessimism-specific. A non-durables placebo does not hold cleanly. The results document substantial belief stickiness in durable-goods expectations but provide only qualified evidence that it translates into measurable spending distortions.

1. Introduction

In a June 2022 newsletter, Economic Commentator Kyla Scanlon (Scanlon, 2022) coined the term "vibecession" describing a contradiction between a country's economic indicators and the general American's attitude towards the economy. Term used later by Treasury Secretary Scott Bessent (Shaban, 2025) in reference to the Biden Economy, where strong macroeconomic indicators could not fully articulate the overall reality of soaring costs. So while economies cyclically enter rough patches, recovery rarely feels as fast as the data suggests it should be. Unemployment falls, incomes rise, credit conditions ease, and still consumers hold back. They delay replacing the aging appliance, postpone buying a new car, put off the furniture they've been eyeing for months. By the time they finally open their wallets, economists have already declared the recession over.

This gap between sentiment and reality has been widely covered in Macroeconomics. One answer, which this paper investigates, is that consumers are slow to update what they believe about the economy. When conditions deteriorate, households hold a pessimistic attitude that tends to linger well past the point where the macroeconomic data warrants it. This sluggishness in belief updating—also referred to as belief stickiness—drives a wedge between what consumers think about the economy and what the economy is actually doing.

This paper studies belief stickiness through the lens of durable goods consumption precisely because it is the spending category where distorted beliefs should show up most clearly. Durable goods—major household appliances, furniture, electronics, automobiles—are optional and postponable in a way that food or electricity is not, making their purchase directly contingent on forward-looking confidence. This discretionary quality makes durable goods spending a natural laboratory for studying how beliefs shape economic behavior.

To measure belief stickiness, I construct two parallel series. The first is a subjective belief series drawn from the University of Michigan Survey of Consumers, specifically the Index of Buying Conditions for Large Household Durables, which asks respondents each month whether they think it is a good time to make major household purchases. This directly captures what consumers believe about economic conditions as they relate to durable spending decisions. The second is a rational forecast benchmark—what an informed observer, armed with the same macroeconomic data available at the time, would have predicted about durable buying conditions. I construct this benchmark using a LASSO regression model that selects among twenty-one real-time macroeconomic predictors including labor market conditions, financial indicators, pricing variables, and measures of output and income. The gap between these two series is my measure of belief distortion. Belief stickiness is one specific mechanism that could give rise to such a distortion, tested separately below.

With this measure in hand, the paper asks a simple question: does it matter? Specifically, does the gap between consumer beliefs and their rational counterpart predict actual durable goods spending over and above what current macro conditions already explain? If belief stickiness is economically meaningful, pessimistic consumers should spend less on durables than fundamentals warrant, and the stickiness measure should have predictive content for future spending at horizons of one, three, and six months.

To validate the measure, I also conduct a natural placebo test built into the structure of the problem. If the stickiness measure is genuinely capturing distortions in discretionary, expectation-sensitive spending, it should predict durable goods consumption but not non-durable goods consumption, because non-durable purchases like food and personal care products are driven by necessity rather than forward-looking beliefs. A finding that stickiness predicts durables but not non-durables would be difficult to explain by any story other than the expectation channel I have in mind.

This paper is most directly related to Du, Monninger, Qiu, and Wang (2025), who study belief stickiness in the context of household unemployment risk perceptions using a similar LASSO-based rational benchmark methodology. They find that subjective job loss and job finding expectations adjust sluggishly to macroeconomic conditions, and that this stickiness attenuates the precautionary saving channel, causing households to under-insure during recessions and recover more slowly. I apply the same methodological framework to consumer durable goods beliefs, asking whether a parallel mechanism operates on the spending side of the household balance sheet. The data are drawn from the University of Michigan Survey of Consumers, spanning January 1997 to December 2024, with macroeconomic predictors sourced from the FRED-MD database of McCracken and Ng (2016).

2. Literature Review

This paper connects to four strands of existing literature, each of which it builds on while identifying a gap it aims to fill.

The first strand provides the theoretical foundations for belief stickiness. Mankiw and Reis (2002) propose a sticky information model in which agents update their full information set only occasionally; each period, a fraction of agents acquire updated macroeconomic information while the rest continue acting on stale beliefs. The result is that aggregate expectations adjust sluggishly to new data even though each individual agent behaves optimally given their information. Sims (2003) provides a complementary microfoundation through the theory of rational inattention: because processing information is cognitively costly, agents optimally choose to be selectively inattentive, rationally devoting limited attention to the macroeconomic signals most relevant to their decisions. Together these papers establish that belief stickiness is not a behavioral failure but an equilibrium outcome of optimal information management under constraints. Most directly relevant to this paper, Carroll, Crawley, Slacalek, Tokuoka, and White (2020) show that sticky macroeconomic expectations generate the well-documented excess smoothness of aggregate consumption growth; households acting on outdated beliefs about the macroeconomy consume as if living in a past economic state, producing sluggishness that rational expectations models cannot match without ad hoc assumptions. This paper takes their theoretical prediction as its empirical premise and tests whether the consumption distortions generated by sticky beliefs are visible and measurable in the component of spending most sensitive to forward-looking expectations: durable goods.

The second strand is the empirical literature on household expectation formation. Coibion and Gorodnichenko (2015) show that household forecasts exhibit systematic and predictable errors consistent with information rigidity; households absorb new information more slowly than a rational agent would.

Carroll (2003) proposes an epidemiological model in which rational information diffuses slowly through the population, generating persistent disagreement and forecast errors. Mankiw, Reis, and Wolfers (2003) document substantial disagreement in inflation expectations across households. Taken together, these papers establish that belief updating is sluggish and that deviations from rationality are systematic rather than random. However, a critical gap remains: while these papers carefully characterize the expectation errors themselves, they do not trace the downstream consequences of those errors for household spending decisions. This paper addresses that gap directly by linking measured belief stickiness to realized durable goods expenditure.

The third strand is the literature connecting household expectations to consumption behavior. Mishkin (1978) shows that the Michigan Index of Consumer Sentiment has good explanatory power for consumer durable expenditure, and Souleles (2004) confirms using household microdata that sentiment forecasts future consumption even after controlling for income and other macroeconomic variables. Carroll (1994) establishes that income uncertainty is a quantitatively important determinant of consumption decisions, grounding the theoretical link between expectations and spending. What this literature does not do is decompose sentiment into its rational and irrational components; it treats survey responses as a unified signal rather than asking how much of that signal reflects genuine information versus sluggish belief updating. That decomposition is the central methodological contribution of this paper.

The fourth and most proximate strand is the growing literature on belief stickiness and its macroeconomic consequences. Du, Monninger, Qiu, and Wang (2025), the paper this work most directly builds upon, study belief stickiness in the context of household unemployment risk perceptions. Using the Survey of Consumer Expectations backcast to 1978 and real-time LASSO forecasting to construct a rational benchmark, they document that subjective job loss and job finding expectations adjust sluggishly to changes in actual labor market conditions. Critically, they show that this stickiness attenuates the precautionary saving channel during recessions: because workers underestimate their unemployment risk, they under-insure and experience a more sluggish consumption recovery. This paper adopts the same methodological framework but applies it to beliefs about durable goods buying conditions rather than unemployment risk. This extension is non-trivial. The labor market channel Du et al. identify operates through precautionary saving; the channel examined here operates through direct discretionary spending decisions. Together the two mechanisms paint a more complete picture of how belief stickiness shapes household economic behavior. A key gap that remains in the literature, and which this paper begins to address, is whether belief stickiness in consumer sentiment indices generates predictable and exploitable distortions in aggregate consumption data.

The remainder of the paper is organized as follows. Section 3 describes the methodology, including the LASSO estimation procedure, the construction of the belief stickiness measure, the predictive regression framework, and the placebo test. Section 4 presents the data, describing the subjective belief series, the macroeconomic predictor set, and the outcome variables. Section 5 reports results. Section 6 discusses limitations. Section 7 concludes.

3. Methodology

The empirical strategy consists of three stages: constructing the subjective belief series, constructing the rational forecast benchmark, and measuring and testing belief stickiness.

3.1 Subjective Belief Series

The subjective belief variable is the University of Michigan Index of Buying Conditions for Large Household Durables (UMSBMQ), which measures the share of survey respondents who report that now is a good time to buy major household goods. This series directly captures consumers' forward-looking assessment of whether economic conditions, including their own income prospects, prevailing prices, and the general economic outlook, are favorable for large discretionary purchases. Let B^s_t denote this subjective belief index at time t. As a robustness check, we also report results using the broader Index of Consumer Sentiment (UMCSENT) and one-year inflation expectations (MICH) as alternative belief measures.

3.2 Rational Forecast Benchmark

The rational forecast benchmark is constructed using a LASSO regression model estimated on a set of N = 21 real-time macroeconomic predictors. Adapting the methodology of Du, Monninger, Qiu, and Wang (2025), the LASSO is trained to predict the realized outcome of interest, real PCE durable goods growth, rather than the subjective belief itself. The fitted value at each month t represents the level of contemporaneous spending growth that a fully informed rational agent would forecast given the available macroeconomic data, and is denoted $\hat{c}_t$.

At each point in time t, the model is re-estimated using only data available at time t. To enforce the real-time constraint, all predictors are lagged by their respective publication delays as described in Section 2.

One might instead train the LASSO directly on the subjective belief series, fitting macroeconomic predictors to the survey response itself. This paper deliberately does not, because doing so would build circularity into the benchmark: a model fit to the belief would partially absorb the very sticky component the benchmark is meant to exclude, contaminating the rational forecast with the sluggishness it is supposed to measure against. Anchoring the benchmark to realized spending growth instead keeps the two series conceptually distinct. The trade-off is that the rational forecast $\hat{c}_t$ is then on the scale of a percentage growth rate while the subjective belief (the UMSBMQ index) is on an index scale; this difference of units is reconciled in Section 3.3 below via standardization.

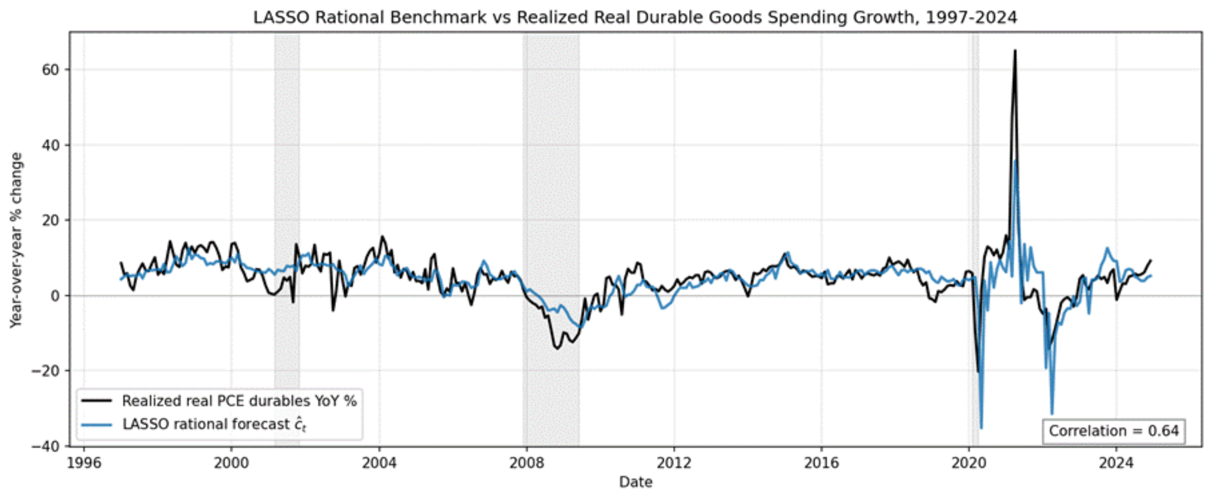

Figure 1: LASSO Rational Benchmark vs Realized Real Durable Goods Spending Growth, 1997-2024. The black line plots realized year-over-year growth in real PCE durable goods (PCEDG deflated by CUSR0000SAD); the blue line plots the rolling-window LASSO forecast $\hat{c}$ *trained on 21 real-time macroeconomic predictors. Shaded regions indicate NBER recession dates.*

3.3 Belief Distortion Measure

This stage constructs a direct measure of the gap between consumers' subjective survey response and the rational, fundamentals-implied forecast of spending. Following the terminology of Du, Monninger, Qiu, and Wang (2025), I refer to this gap as the belief distortion: the deviation of perceived buying conditions from the level a fully informed observer would forecast given the available real-time macroeconomic data. Belief stickiness, by contrast, refers to a specific hypothesis about why such distortions arise, namely that subjective beliefs update sluggishly to new fundamentals. Stickiness is one of several mechanisms that could in principle generate the observed distortion, alongside private information held by survey respondents, signal-extraction frictions, and rational inattention. Section 3.4 isolates the stickiness mechanism by testing it directly via the belief-elasticity regression of Du et al. (2025, Eq. 6).

Because the subjective belief (the UMSBMQ index, ranging approximately from 50 to 180) and the LASSO rational forecast (a percentage growth rate) are measured on different scales, I standardize each series to z-scores over the sample period and define the belief distortion measure as:

$$D_t = z(B_t^s) - z(\hat{c}_t)$$

where z(x) denotes standardization. A positive D_t indicates that consumers are more optimistic about durable buying conditions than the fundamentals-implied forecast of spending would warrant; a negative D_t indicates that consumers are more pessimistic, the economically consequential case under the stickiness hypothesis, in which households delay durable purchases despite improving macroeconomic fundamentals.

Standardization provides a common scale on which the distortion can be visualized and characterized over the business cycle. Two caveats follow. First, because the subjective index and the rational growth-rate forecast are not in natural common units, D_t is the difference of two arbitrary standardizations rather than a gap with a direct economic interpretation; its level and scale carry no structural meaning. Second, the persistence of D_t is partly mechanical, since both underlying series are themselves persistent. D_t is therefore used in this paper only as a descriptive device for tracking belief-fundamentals divergence over time. The formal evidence on whether the observed distortion reflects belief stickiness specifically comes from the elasticity test of Section 3.4 and the predictive regressions of Section 3.5, in which the subjective belief and the rational forecast enter as separate regressors rather than as a constructed gap.

3.4 Testing the Macroeconomic Consequences of Belief Stickiness

The formal test of whether belief stickiness has measurable consequences for actual spending is conducted via the following predictive regression at forecast horizons h = {1, 3, 6} months:

$$\Delta c_{t+h} = \alpha + \beta_1 \hat{c}_t + \beta_2 {B^s}_t + \gamma' X_t + \varepsilon_{t+h}$$

where Δc_{t+h} is the year-over-year growth rate of real PCE durable goods spending at time t+h; $\hat{c}_t$ is the LASSO rational forecast of contemporaneous spending growth constructed in Section 3.2; ${B^s}_t$ is the subjective survey belief (the UMSBMQ index); and X_t is a vector of macroeconomic controls comprising the unemployment rate (income channel), headline CPI year-over-year inflation (purchasing-power channel), the 10-year Treasury yield (financing-cost channel), and the lagged outcome y_t. The lagged outcome captures the autoregressive persistence in spending growth, isolating the marginal predictive contribution of beliefs over and above the outcome's own momentum.

The coefficient of interest is β_2. If consumer beliefs were perfectly rational and reflected only macroeconomic fundamentals, ${B^s}_t$ would be redundant once we control for $\hat{c}_t$ and the other macro variables, implying $\beta_2 = 0$. A significant $\beta_2 \neq 0$ indicates that the subjective belief carries behaviorally relevant information beyond what the rational macro forecast already captures, that is, that sticky beliefs translate into measurable deviations of actual spending from the rational benchmark.

This paper also delves to check if pessimism prolongs consumption recoveries: households are reluctant to make discretionary durable purchases when they perceive conditions to be worse than fundamentals warrant, but optimistic deviations do not symmetrically accelerate spending. To test this prediction directly, we split the subjective belief into deviations above and below its sample mean μ:

$$\Delta c_{t+h} = \alpha + \beta_1 \hat{c}_t + \beta^-({B^s}_t - \mu)^- + \beta^+({B^s}_t - \mu)^+ + \gamma' X_t + \varepsilon_{t+h}$$

where $({B^s}_t - \mu)^- = min({B^s}_t - \mu, 0)$ captures pessimistic deviations and $({B^s}_t - \mu)^+ = max({B^s}_t - \mu, 0)$ captures optimistic deviations. The theoretical prediction is $\beta^- < 0$ (pessimistic beliefs predict weaker future spending) and $\beta^+ \approx 0$ (optimistic deviations do not symmetrically boost spending). Standard errors in both specifications are computed using the Newey-West heteroskedasticity and autocorrelation consistent estimator with a bandwidth of h lags, accounting for the serial correlation induced by overlapping forecast horizons at h > 1.

3.5 Placebo Test

As a validation of the identification strategy, we re-estimate the predictive regressions of Section 3.4 replacing the real PCE durable goods growth outcome with the year-over-year growth rate of real PCE non-durable goods spending. Non-durable purchases such as food, energy, clothing, and personal care are driven primarily by necessity rather than discretionary forward-looking expectations, and should therefore be largely insensitive to distortions in beliefs about durable buying conditions.

A finding that the belief coefficient β_2 (in the symmetric specification) or β^- (in the asymmetric specification) is significant for durables but insignificant for non-durables would provide strong validation that the belief variable is capturing genuine expectation-driven distortions in discretionary spending, rather than reflecting a general omitted variable correlated with both beliefs and all categories of consumption.

3.6 Rolling Window Estimation

Following the real-time spirit of Du, Monninger, Qiu, and Wang (2025), the LASSO model is estimated on a rolling ten-year window of data rather than using the full sample. At each month t, the model is re-estimated using data from $t-120$ to $t-1$, and the resulting fitted value is used as $\hat{c}_t$ for that month only. The use of training data strictly prior to t enforces a genuine out-of-sample interpretation: the rational benchmark at time t reflects only information that was available before the LASSO model was used to produce the forecast for t.

This rolling procedure ensures that the rational benchmark reflects only information that was genuinely available and learnable at time t, and that structural changes in the relationship between macroeconomic variables and durable goods spending, like those induced by the 2008 financial crisis or the COVID-19 pandemic, do not contaminate the out-of-sample flavor of the exercise. The penalty parameter λ_t is re-selected by time-series cross-validation at each rolling window.

4. Data

This study draws on monthly U.S. data spanning January 1997 to December 2024, a sample of 336 monthly observations. The subjective consumer belief is measured by the University of Michigan Index of Buying Conditions for Large Household Durables (UMSBMQ), obtained from the Surveys of Consumers via FRED. The index summarizes respondents' assessment of whether now is a good or bad time is to make major household purchases and ranges approximately from 50 to 180 over the sample. Its use as the primary belief variable is motivated by Mishkin (1978), who showed that Michigan Survey responses have good explanatory power for durable expenditure, and Souleles (2004), who confirmed that consumer sentiment forecasts future consumption even after controlling for macroeconomic variables. The focus on durables reflects the observation that such purchases are discretionary and easily postponed, making them the domain where forward-looking beliefs matter most.

The objective macroeconomic predictor-set comprises 21 real-time indicators following the spirit of McCracken and Ng (2016). Table 1 summarizes each variable, its transformation, publication lag, and the rationale for inclusion where non-standard.

Table 1: Macroeconomic Predictor Set

Variable	**FRED Code**	**Transformation**	**Lag**	**Notes**
Unemployment rate	UNRATE	Levels	1 month	
Industrial production	INDPRO	YoY % change	1 month	
Manufacturing employment	MANEMP	YoY % change	1 month	
Real GDP growth	A191RL1Q225SBEA	Levels	3 months	Quarterly; forward-filled to monthly
Durable goods orders	DGORDER	YoY % change	2 months	
Retail sales	RSXFS	YoY % change	1 month	
Housing starts	HOUST	Levels	1 month	
10-year Treasury yield	DGS10	Levels	None	Available in real time
Federal funds rate	FEDFUNDS	Levels	None	Available in real time
10-year minus 2-year spread	T10Y2Y	Levels	None	Estrella and Mishkin (1996)
Personal savings rate	PSAVERT	Levels	1 month	
Headline CPI	CPIAUCSL	YoY % change	1 month	
Core CPI	CPILFESL	YoY % change	1 month	
PCE price index	PCEPI	YoY % change	1 month	
CPI for durables	CUSR0000SAD	YoY % change	1 month	

PCE durables price deflator	DDURRD3Q086SBEA	YoY % change	3 months	Quarterly; forward-filled to monthly
PPI final intermediate supply	PPIFIS	YoY % change	1 month	Begins late 2009; dropped from earlier rolling windows automatically
Import price index, mfg durables ex auto	IR41	YoY % change	1 month	McCarthy (2000)
Real disposable personal income	DSPIC96	YoY % change	1 month	Carroll (1994)
Consumer sentiment	UMCSENT	Levels	None	Carries forward-looking information not yet in lagged aggregates
1-year inflation expectations	MICH	Levels	None	Carries forward-looking information not yet in lagged aggregates

The primary outcome variable is the year-over-year growth rate of real PCE durable goods spending. Because FRED's official chain-weighted real PCE durables series (PCEDGC96) begins only in 2007, I construct a real durables series for the full sample by deflating nominal PCE durable goods (PCEDG) by the CPI for durables (CUSR0000SAD):

$$real\_PCEDG_t = \left(\frac{PCEDG_t}{CUSR0000SAD_t}\right) \times 100$$

The year-over-year percent change of this series serves as the LASSO target and the dependent variable in the predictive regressions. The placebo outcome is the year-over-year growth rate of PCE non-durable goods (PCND), available at quarterly frequency and forward-filled to monthly.

In addition to the full sample, results are reported across two trimmed samples to assess robustness. The No COVID sample drops five months affected by the pandemic shock and reopening base effects (May to June 2020 and April to June 2021). The No Abnormal sample drops fourteen months in which the unemployment rate, year-over-year industrial production growth, or year-over-year retail sales growth exhibits a *z*-score exceeding three standard deviations in absolute value; these flagged months span December 2008 through August 2009, May to June 2020, and April to June 2021.

5. Results

5.1 Distortion as a Time Series

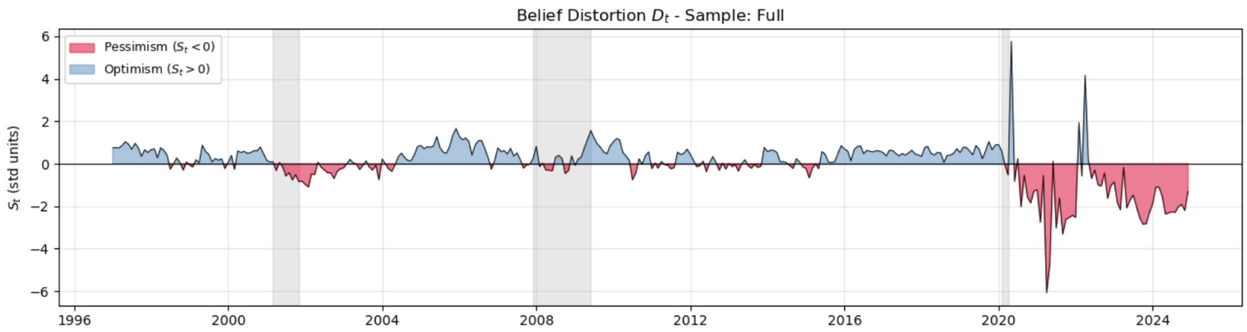

Fig 1: Belief Distortion D_t for the full sample, January 1997–December 2024. Blue regions ($D_t > 0$) denote optimism relative to fundamentals; red regions ($D_t < 0$) denote pessimism. Shaded vertical bands mark NBER recessions.

Figure [1] plots the standardized belief stickiness measure $S_t = z(B_t^s) - z(\hat{c}_t)$ from January 1997 to December 2024 with NBER recession periods shaded. The series exhibits some moderate persistence as the AR(1) coefficient is approximately 0.60 in the full sample and similar in the trimmed samples, implying a half-life of roughly one and a half months. This degree of persistence is consistent with the sticky-information mechanism of Mankiw and Reis (2002), in which only a fraction of households update their information set in any given period, generating gradual rather than instantaneous adjustment of aggregate beliefs to fundamentals. Stickiness reaches its most negative values during the 2008–09 financial crisis and the early COVID-19 period, indicating that consumer beliefs were more pessimistic than the fundamentals-implied benchmark would warrant during both episodes; symmetrically, S_t turns sharply positive in the 2021 reopening period, when optimism about durable buying conditions outran what fundamentals justified.

Table 2: Summary Statistics for Standardized Distortion $D_t = z(B_t^s) - z(\hat{c}_t)$

Sample	N	Mean	Std	Min	Max	AR(1)	Half-life	$corr(B, \hat{c})$
Full	336	−0.00	1.06	−6.08	+5.76	0.624	1.47	0.443
No COVID	331	+0.00	0.89	−3.86	+4.14	0.640	1.56	0.600
No abnormal	322	+0.00	0.92	−4.08	+4.69	0.587	1.30	0.576

Summary statistics for S_t are reported in Table [2]. The mean is approximately zero by construction, the unconditional standard deviation is approximately one, and the range spans roughly $[-3, +3]$ standard deviations. The trimmed samples display modestly lower volatility and similar persistence, indicating that the basic time-series properties of stickiness are not driven by the small number of extreme-shock months identified by the $|z| > 3$ outlier rule. The formal statistical test of whether belief stickiness has measurable consequences for spending is conducted via the predictive regression in Section 3.4 below, in which both the rational forecast and the subjective belief enter as separate regressors.

5.2 First-Stage Belief Elasticity Test

Following Du, Monninger, Qiu, and Wang (2025, Eq. 6), I regress the standardized subjective belief on the standardized rational forecast in each sample and test the null of rational expectations, H_0: $\beta = 1$, against the alternative of underreaction, H_1: $\beta < 1$. Table 3 reports the estimates with Newey-West HAC standard errors and the one-sided p-value for the rationality null.

The rationality null is rejected at the 0.01% level in every sample, suggesting that subjective beliefs about durable buying conditions are statistically distinguishable from a rational expectations benchmark by a wide margin.

The point estimates indicate that in the full sample, a one-standard-deviation movement in the rational forecast translates into only a 0.443-standard-deviation movement in subjective beliefs, i.e. beliefs adjust by roughly 44% of what fundamentals warrant. In the two trimmed samples that exclude the most extreme macroeconomic shocks, the elasticity rises to approximately 0.58 to 0.60, indicating that in non-shock periods consumer beliefs are somewhat less sluggish but still adjust by only about 60% of the rational benchmark. The headline durables stickiness estimate of 0.58 in the No Abnormal sample is strikingly close to the value of 0.51 that Du, Monninger, Qiu, and Wang (2025) report for perceived unemployment risk in their Equation 6, suggesting that the degree of belief stickiness in consumer expectations is comparable across distinct economic domains.

The pattern across samples carries its own information. The elasticity is lower in the full sample than in either trimmed sample, with the largest gap relative to rationality observed when shock periods are included. One interpretation is that during periods of extreme macroeconomic dislocation, the 2008–09 financial crisis trough, the COVID-19 shock, and the 2021 reopening base-effect period, rational fundamentals move rapidly while consumer beliefs adjust even more slowly than in normal times. Belief stickiness thus appears not only to exist on average but to intensify precisely when sluggish updating is most consequential for the macroeconomy. The accompanying rise in regression R^2 from 0.196 in the full sample to 0.33-0.36 in the trimmed samples is consistent with this interpretation: extreme-shock observations add noise to the belief–fundamentals relationship without altering the underlying conclusion that beliefs underreact.

Together with the persistence of S_t documented in Section 5.1 (AR(1) ≈ 0.60), this first-stage result establishes that the belief stickiness studied in this paper is statistically detectable, economically substantial, and quantitatively comparable to the stickiness Du et al. document in the labor market. Section 5.3 next turns to the central question of whether this stickiness has measurable consequences for actual durable spending.

Table 3: Du-Style Underreaction Test: First-Stage Belief Elasticity

	$z(UMSBMQ_t) = \alpha + \beta \cdot z(\hat{c}_t) + \varepsilon$					
Sample	$\hat{\beta}$	SE	$t(\beta = 1)$	p (one-sided)	R^2	N
Full	+0.443	0.109	−5.12	0.0000***	0.196	336
No COVID	+0.600	0.072	−5.58	0.0000***	0.360	331
No abnormal	+0.576	0.078	−5.44	0.0000***	0.332	322

5.3 Consequences for Real Durable Spending: Symmetric Specification

Table [4] reports the main predictive regression of equation

$$\Delta c_{t+h} = \alpha + \beta_1 \hat{c}_t + \beta_2 B_t^s + \gamma' X_t + \phi \Delta c_t + \varepsilon,$$

in which real PCE durable goods growth at horizon h is regressed on the rational forecast $\hat{c}_t$, the subjective belief B_t^s, and macroeconomic controls. The coefficient of interest is $\beta_2$on B_t^s, which under the joint inclusion of $\hat{c}_t$measures the marginal predictive content of the residual, sticky component of beliefs.

The symmetric specification does not, on its own, yield significant evidence that stickiness predicts subsequent durable spending. Across all three samples and three horizons (nine cells), $\hat{\beta}_2$is small in magnitude ($|\hat{\beta}_2| < 0.025$ throughout) and statistically insignificant at conventional levels, with two-sided p-values ranging from 0.19 to 0.97. Signs are inconsistent across horizons. The coefficient on the rational forecast, β_1, is similarly small in the symmetric specification, indicating that once unemployment, inflation, the long Treasury yield, and the lagged outcome are included as controls, neither the LASSO-implied rational forecast nor the subjective survey response carries additional predictive content for durable spending growth at the average month.

Table 4: Symmetric Consequences Regression: Dependent Variable: Real PCE Durable Goods Growth

$\Delta c_{t+h} = \alpha + \beta_1 \hat{c}_t + \beta_2 B_t^s + \gamma' X_t + \phi \Delta c_t + \varepsilon$

	Full			No COVID			No abnormal		
	$h=1$	$h=3$	$h=6$	$h=1$	$h=3$	$h=6$	$h=1$	$h=3$	$h=6$
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
UMSBMQ ($\hat{\beta}_2$)	−0.003	+0.004	+0.015	−0.010	+0.005	+0.022	−0.021	−0.013	+0.001
	(0.019)	(0.031)	(0.035)	(0.016)	(0.021)	(0.028)	(0.016)	(0.020)	(0.024)
$\hat{c}_t$ ($\hat{\beta}_1$)	−0.083	+0.004	−0.029	+0.443*	+0.303**	+0.070	+0.330	+0.147	−0.096
	(0.136)	(0.094)	(0.117)	(0.248)	(0.131)	(0.129)	(0.264)	(0.132)	(0.126)
Unemployment rate	−0.160	−0.157	+0.036	+0.189	+0.085	+0.042	+0.020	−0.104	−0.086
	(0.219)	(0.229)	(0.283)	(0.241)	(0.238)	(0.272)	(0.219)	(0.211)	(0.258)
CPI YoY	−0.586**	−0.870*	−0.737	−0.201	−0.348	−0.452	−0.696	−1.053**	−1.117**
	(0.283)	(0.507)	(0.470)	(0.284)	(0.372)	(0.367)	(0.445)	(0.487)	(0.478)
10Y Treasury	+0.154	+0.112	+0.130	+0.085	+0.117	+0.241	+0.239	+0.391	+0.598
	(0.218)	(0.393)	(0.476)	(0.207)	(0.290)	(0.391)	(0.265)	(0.321)	(0.385)
Lagged outcome	+0.777***	+0.372**	+0.296	+0.471***	+0.332**	+0.294*	+0.419**	+0.235*	+0.191
	(0.131)	(0.186)	(0.185)	(0.179)	(0.146)	(0.153)	(0.173)	(0.127)	(0.126)
Constant	+3.707	+4.957	+2.560	+1.017	+0.952	−0.246	+5.061	+6.759	+5.444
	(3.686)	(6.483)	(6.554)	(3.444)	(4.572)	(5.579)	(3.829)	(4.424)	(5.119)
R^2	0.604	0.231	0.163	0.530	0.327	0.218	0.469	0.264	0.174
N	335	333	330	330	328	325	321	319	316

Notes: Dependent variable is real PCE durable goods growth (Δc_{t+h}) at horizon h months. The coefficient of interest $\hat{\beta}_2$ on UMSBMQ measures the predictive content of the sticky component of beliefs, conditional on the LASSO rational forecast $\hat{c}_t$ and macro controls. Standard errors (in parentheses) are Newey-West HAC with maxlags = h. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$ (two-sided). Sample period: January 1997 – December 2024.

The coefficient on the rational forecast $\hat{c}_t$in the symmetric specification (β_chat) is positive and statistically significant at h=1 in the No COVID sample (β = 0.44, p ≈ 0.07) and at h=3 (β = 0.30, p ≈ 0.02). This is a sanity check: it confirms that the LASSO-implied rational forecast carries genuine predictive content for actual spending, which is a non-trivial validation of the benchmark itself.

5.4 Asymmetric Specification

Splitting B_t^s into pessimistic and optimistic deviations from the sample mean of USBMQ yields a more nuanced picture, reported in Table [6]. At the one-month horizon in the No abnormal sample — the most aggressive outlier-trimmed specification, which drops the 14 most extreme macroeconomic months — the pessimism coefficient is $\hat{\beta}^{-} = -0.076$with a two-sided p-value of 0.020, and the optimism coefficient is $\hat{\beta}^{+} = +0.094$with p-value 0.021, both statistically significant in the directions the theory predicts. A milder version of the same pattern is visible in the No COVID sample at the same horizon ($\hat{\beta}^{-} = -0.054, p = 0.087; \hat{\beta}^{+} = +0.081, p = 0.047$). At horizons of three and six months in all samples, and in the full sample at all horizons, neither asymmetric coefficient reaches conventional significance for durables.

Two qualifications follow. First, the durables result is concentrated at the one-month horizon and emerges only in samples that exclude shock months. Second, the magnitudes of $\hat{\beta}^{-}$ and $\hat{\beta}^{+}$are comparable in absolute value where both are significant, the relationship is therefore not pessimism-asymmetric in the strict sense the theory predicts, but is rather a symmetric short-horizon predictive relationship between belief deviations from the mean and durable spending. The data thus provide qualified support for an expectations channel into durable spending at short horizons in non-shock periods, without supporting the specific prediction that pessimism alone drives the effect.

Table 6: Asymmetric Consequences Regression: Dependent Variable: PCE Durable Goods Growth

	$\Delta c_{t+h} = \alpha + \beta_1 \hat{c}_t + \beta^-(B_t^s - \mu)^- + \beta^+(B_t^s - \mu)^+ + \gamma' X_t + \phi \Delta c_t + \varepsilon$								
	Full			No COVID			No abnormal		
	$h = 1$	$h = 3$	$h = 6$	$h = 1$	$h = 3$	$h = 6$	$h = 1$	$h = 3$	$h = 6$
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
UMSBMQ$^-$ (β^-)	−0.019	−0.003	−0.021	−0.054*	+0.002	+0.003	−0.076**	−0.032	−0.033
	(0.026)	(0.042)	(0.063)	(0.032)	(0.031)	(0.045)	(0.033)	(0.030)	(0.042)
UMSBMQ$^+$ (β^+)	+0.030	+0.019	+0.084	+0.081**	+0.011	+0.058	+0.094**	+0.026	+0.065
	(0.036)	(0.052)	(0.075)	(0.041)	(0.046)	(0.068)	(0.041)	(0.046)	(0.068)
$\hat{c}_t$ (β_1)	−0.067	+0.011	+0.005	+0.502*	+0.307**	+0.095	+0.385	+0.165	−0.064
	(0.142)	(0.094)	(0.136)	(0.258)	(0.134)	(0.130)	(0.263)	(0.132)	(0.129)
Unemployment rate	−0.061	−0.114	+0.238	+0.492	+0.105	+0.159	+0.362	+0.010	+0.097
	(0.266)	(0.228)	(0.388)	(0.311)	(0.256)	(0.323)	(0.269)	(0.229)	(0.311)
CPI YoY	−0.590**	−0.873*	−0.763	−0.198	−0.349	−0.460	−0.778*	−1.086**	−1.190**
	(0.286)	(0.514)	(0.495)	(0.290)	(0.373)	(0.373)	(0.450)	(0.500)	(0.489)
10Y Treasury	+0.129	+0.102	+0.085	+0.008	+0.112	+0.215	+0.168	+0.369	+0.570
	(0.212)	(0.398)	(0.501)	(0.204)	(0.295)	(0.408)	(0.261)	(0.327)	(0.401)
Lagged outcome	+0.770***	+0.370**	+0.284	+0.451**	+0.331**	+0.286*	+0.388**	+0.224*	+0.174
	(0.134)	(0.187)	(0.181)	(0.178)	(0.147)	(0.150)	(0.169)	(0.125)	(0.119)
Constant	+2.185	+5.128*	+2.423	−3.440	+1.456	+1.658	−1.220	+3.803	+3.750
	(2.839)	(3.036)	(3.451)	(3.080)	(2.924)	(3.478)	(2.863)	(2.759)	(3.395)
R^2	0.605	0.231	0.169	0.543	0.327	0.220	0.493	0.267	0.181
N	335	333	330	330	328	325	321	319	316

Notes: Dependent variable is PCE durable goods growth (YoY) at horizon h months. The belief is split into pessimistic deviations $(B_t^s - \mu)^-$ and optimistic deviations $(B_t^s - \mu)^+$. Standard errors (in parentheses) are Newey–West HAC with maxlags = h. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$ (two-sided). Sample period: January 1997 – December 2024.

5.5 Non-Durables Placebo

The placebo regression replaces real PCE durable goods growth with year-over-year PCE non-durable goods growth as the dependent variable, holding the right-hand side identical to the main specification. Under the identifying assumption that durable-belief stickiness affects only discretionary, expectation-sensitive spending, the coefficient on B^s_t should be insignificant when non-durables are on the left-hand side, since non-durables include daily items like food that cannot be deferred (Black & Cusbert, 2010).

Table [5] reports the placebo results. Contrary to the identifying assumption, the symmetric placebo coefficient on B^s_t is negative and statistically significant at the three- and six-month horizons across all three samples, with point estimates from −0.026 to −0.043 and p-values as low as 0.002 in the trimmed samples.

The placebo therefore does not hold cleanly. Three interpretations are possible. First, PCE non-durables may not be a clean placebo. The aggregate mixes genuinely non-discretionary items (food at home, energy, household supplies) with discretionary ones (food services, recreation, personal care). Sticky beliefs about durable buying conditions almost certainly correlate with broader household financial outlook, which would also reduce spending on restaurant meals and discretionary travel. The significant pessimism coefficient on non-durables is thus consistent with the sticky-pessimism mechanism operating across multiple discretionary margins, rather than with contamination by an omitted macroeconomic factor.

Second, PCND is quarterly data forward-filled to monthly, which injects artificial serial correlation and may be partly generating the significant placebo coefficient.

Third, the controls (unemployment, inflation, the long-term interest rate, and lagged spending) may not fully capture the macroeconomic information that UMSBMQ carries. In that case the placebo significance would reflect UMSBMQ picking up leftover business-cycle variation rather than a genuine belief-stickiness effect. Addressing this would require richer controls or an instrumental-variables approach.

Table 5: Symmetric Consequences Regression: Dependent Variable: PCE Non-Durable Goods Growth (Placebo)

$$\Delta c_{t+h} = \alpha + \beta_1 \hat{c}_t + \beta_2 B^s_t + \gamma' X_t + \phi \Delta c_t + \varepsilon$$

	Full			No COVID			No abnormal		
	$h=1$	$h=3$	$h=6$	$h=1$	$h=3$	$h=6$	$h=1$	$h=3$	$h=6$
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
UMSBMQ ($\hat{\beta}_2$)	−0.003	−0.026**	−0.040**	−0.007*	−0.029***	−0.043***	−0.010**	−0.028***	−0.036**
	(0.004)	(0.012)	(0.016)	(0.004)	(0.009)	(0.017)	(0.004)	(0.009)	(0.018)
$\hat{c}_t$ ($\hat{\beta}_1$)	+0.023	−0.060	−0.003	−0.011	−0.004	+0.025	−0.015	+0.037	+0.114
	(0.027)	(0.040)	(0.047)	(0.020)	(0.050)	(0.088)	(0.024)	(0.050)	(0.084)
Unemployment rate	+0.061	−0.161	−0.122	−0.062	−0.213**	−0.223*	−0.057	−0.127	−0.047
	(0.089)	(0.133)	(0.146)	(0.054)	(0.087)	(0.121)	(0.056)	(0.092)	(0.167)
CPI YoY	−0.170	−1.140***	−1.589***	−0.328***	−1.125***	−1.679***	−0.380***	−0.812***	−0.968**
	(0.165)	(0.327)	(0.313)	(0.094)	(0.224)	(0.319)	(0.110)	(0.170)	(0.394)
10Y Treasury	+0.021	+0.125	+0.139	+0.026	+0.061	+0.117	+0.050	+0.034	+0.012
	(0.059)	(0.128)	(0.195)	(0.053)	(0.111)	(0.186)	(0.062)	(0.118)	(0.185)
Lagged outcome	+0.975***	+1.094***	+1.011***	+1.027***	+1.147***	+1.069***	+1.011***	+0.998***	+0.867***
	(0.063)	(0.107)	(0.090)	(0.031)	(0.065)	(0.113)	(0.037)	(0.090)	(0.168)
Constant	+0.461	+6.855**	+9.746***	+1.984**	+7.194***	+10.745***	+2.531**	+6.440***	+7.949**
	(1.277)	(2.741)	(3.259)	(0.928)	(1.891)	(3.147)	(1.022)	(1.740)	(3.644)
R^2	0.862	0.662	0.488	0.870	0.734	0.501	0.830	0.635	0.442
N	335	333	330	330	328	325	321	319	316

Notes: Dependent variable is PCE non-durable goods growth (YoY) at horizon h months. The right-hand side is identical to Table 4. Under the identifying assumption that durable-belief stickiness is category-specific, $\hat{\beta}_2$ on UMSBMQ should be insignificant here. Standard errors (in parentheses) are Newey-West HAC with maxlags = h. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$ (two-sided). Sample period: January 1997 – December 2024.

6. Limitations

This paper's empirical strategy faces three principal limitations that warrant acknowledgement before drawing strong conclusions from the results.

First, the analysis is conducted on aggregate monthly time-series data, while the underlying behavioral mechanism of belief stickiness operates at the household level. A consumer who reports pessimism about durable buying conditions may be responding optimally to private information about her own income prospects, employment security, or financing constraints that the 21 aggregate macroeconomic predictors in the LASSO benchmark cannot observe. The gap between subjective beliefs and the fundamentals-implied forecast may therefore reflect rational private information at the individual level rather than sluggish belief updating at the population level, and the two explanations cannot be cleanly separated without household-level panel data on both beliefs and spending decisions.

Second, the asymmetric specification of Section 5.4 yields only a single conventionally significant pessimism coefficient, at the one-month horizon in the No Abnormal sample, across eighteen cells. The asymmetric finding is therefore reported as suggestive rather than confirmatory, and the paper's central conclusions rest more securely on the systematic first-stage underreaction result of Section 5.2.

Third, the non-durables placebo is imperfect because PCE non-durables mixes genuine necessities with discretionary categories such as food services and recreation that likely respond to the same belief-formation channel as durables. The significant placebo coefficient in Section 5.5 therefore weakens but does not invalidate the durables-specific interpretation; re-estimating the placebo on necessity-only non-durables would provide a sharper test and is a natural extension.

7. Conclusion

This paper asked whether sticky consumer beliefs about durable buying conditions distort actual durable spending beyond what macroeconomic fundamentals explain. Adapting the rational-benchmark methodology of Du, Monninger, Qiu, and Wang (2025) to a setting where the subjective belief has no directly comparable realized counterpart, I built a rolling-window LASSO forecast of real PCE durable goods growth and compared it against the University of Michigan Index of Buying Conditions for Large Household Durables.

The evidence arrives in three tiers of decreasing strength. First, and most robustly, consumer beliefs underreact to fundamentals: the rationality null is rejected decisively in every sample, with beliefs adjusting by only roughly 44 to 60 percent of what the rational forecast warrants, a degree of stickiness strikingly close to what Du et al. document for unemployment-risk perceptions. That belief stickiness exists in the durables domain, and is quantitatively comparable to stickiness in the labor market, is the paper's firmest result. Second, the spending consequences of this stickiness are weaker and more fragile than the theory would predict. The symmetric predictive regression finds no significant effect of beliefs on subsequent durable spending, and the asymmetric specification yields a significant short-horizon effect only in shock-trimmed samples, with optimistic and pessimistic deviations of comparable magnitude rather than the pessimism-specific asymmetry the mechanism implies. The data therefore support an expectations channel into durable spending only at short horizons and in non-shock periods, and only suggestively. Third, the non-durables placebo does not hold cleanly, which prevents a sharp identification of the channel as durables-specific.

Taken together, the results establish that belief stickiness in durable-goods expectations is real, substantial, and measurable, while showing that its translation into aggregate spending distortions is harder to pin down with reduced-form time-series tools than the underlying theory suggests. This pattern is itself informative: it points to the gap between documenting sticky beliefs and demonstrating their macroeconomic bite as the place where the more demanding work remains. Household-level panel data linking beliefs to spending would separate sluggish updating from rational private information; a necessity-only placebo would sharpen the identification; and a structural treatment in the spirit of Du et al. could quantify the consumption distortion that the reduced-form regressions here can only gesture toward. What this paper contributes is a clear, replicable measurement of stickiness on the spending side of the household balance sheet, and an honest account of how far that measurement does and does not carry.

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